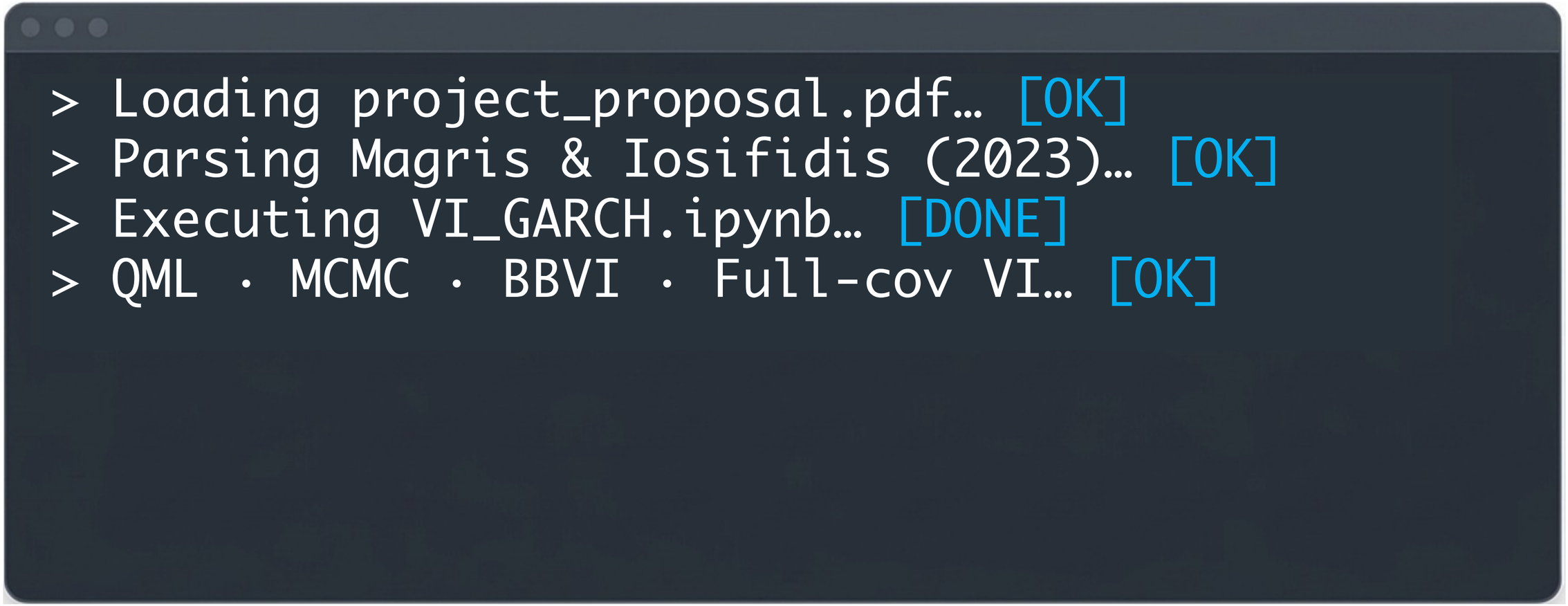
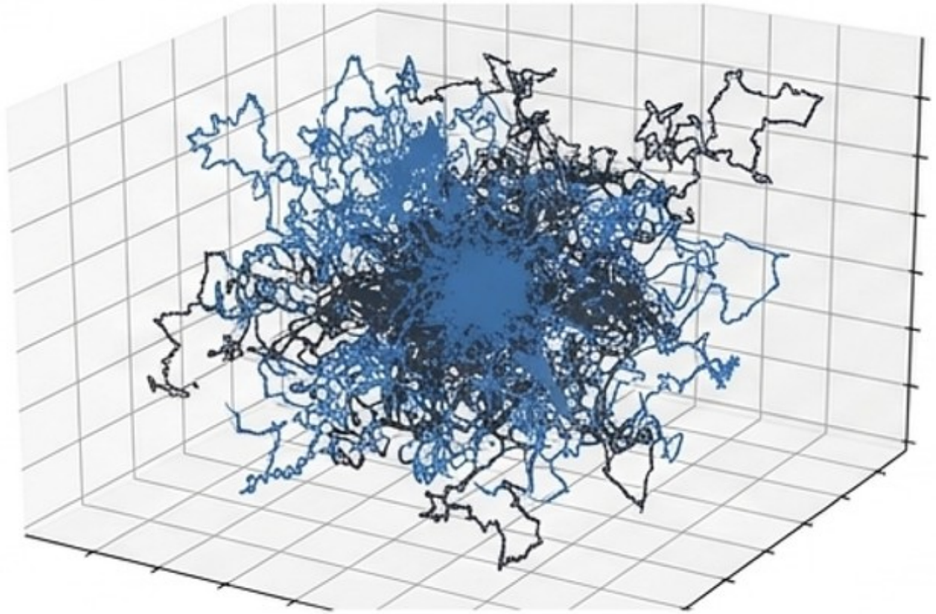
# Bayesian Volatility at Scale




Runtime: NumPy + PyTorch | Data: MSFT | Method: VI

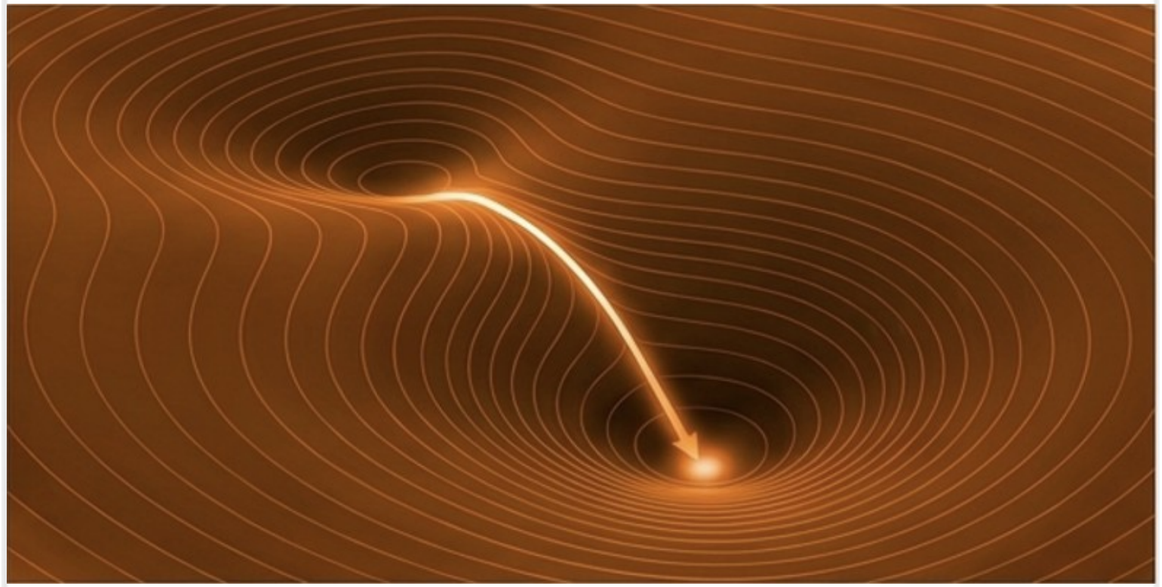
# MCMC is the Bayesian gold standard, but it becomes a bottleneck at scale

## Traditional Bayesian Estimation (Markov Chain Monte Carlo)



- **Method**: Random-walk posterior sampling
- **Cost**: Sequential draws, tuning, diagnostics
- **Scalability**: Mixes slowly in high dimensions

## The Chosen Architecture (Variational Inference)



- **Method**: Optimize the ELBO (momentum / Adam)
- **Cost**: One convergence curve, generic tuning
- **Scalability**: Built for high-dimensional models

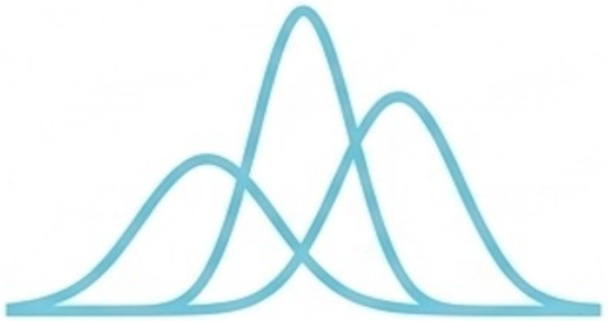
# Variational Inference recasts Bayesian estimation into a scalable optimization problem

## 1. The Intractable True Posterior



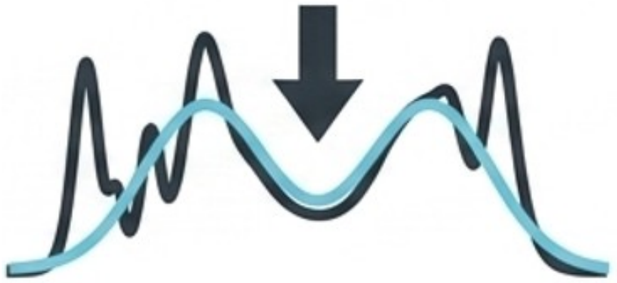
Marginal likelihood **p(y)** is often intractable.

## 2. The Variational Family



A family of tractable proxy distributions $\boldsymbol{q(\theta)}$ parameterized by $\boldsymbol{\zeta}$.

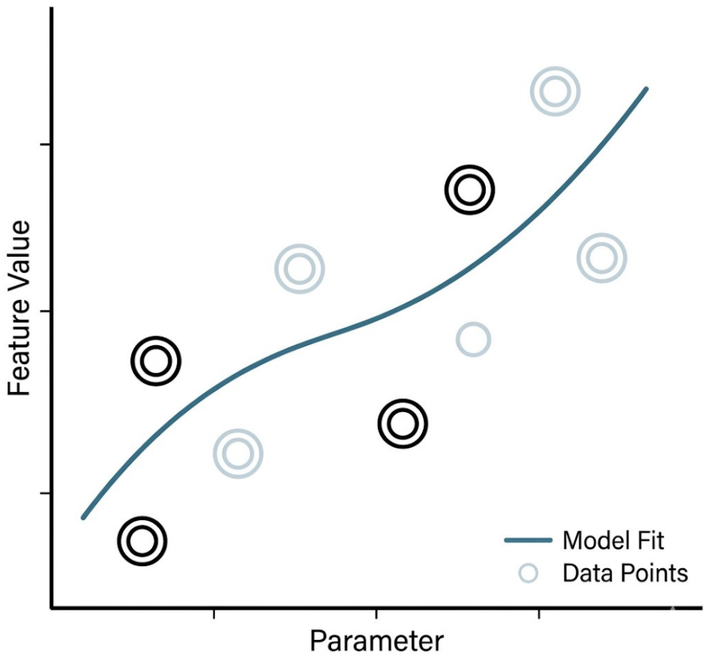
## 3. The Optimization Bridge



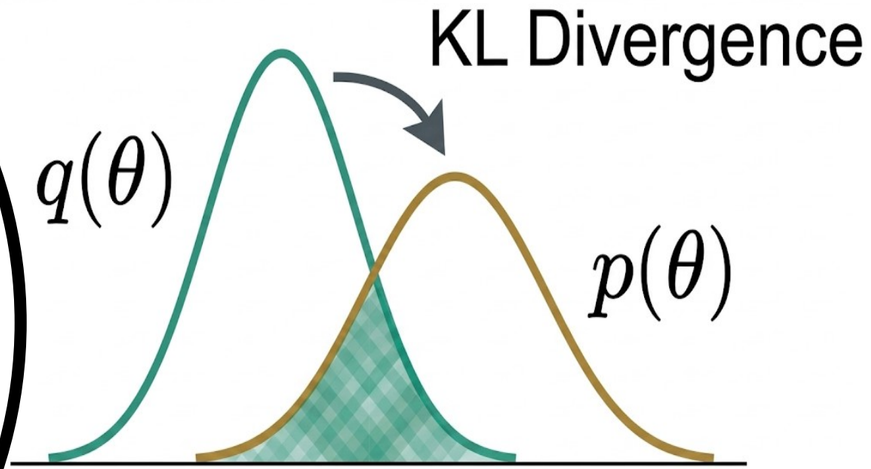
Minimize the Kullback-Leibler (KL) Divergence to force the proxy $\boldsymbol{q(\theta)}$ to match the true posterior.

# The Evidence Lower Bound (ELBO) objective balances data fit against prior beliefs

*The Expected Log-Likelihood*




$$ELBO = \mathbb{E}[log\ p(data \mid \theta)] - \mathrm{KL}(q(\theta) \,||\, p(\theta))$$



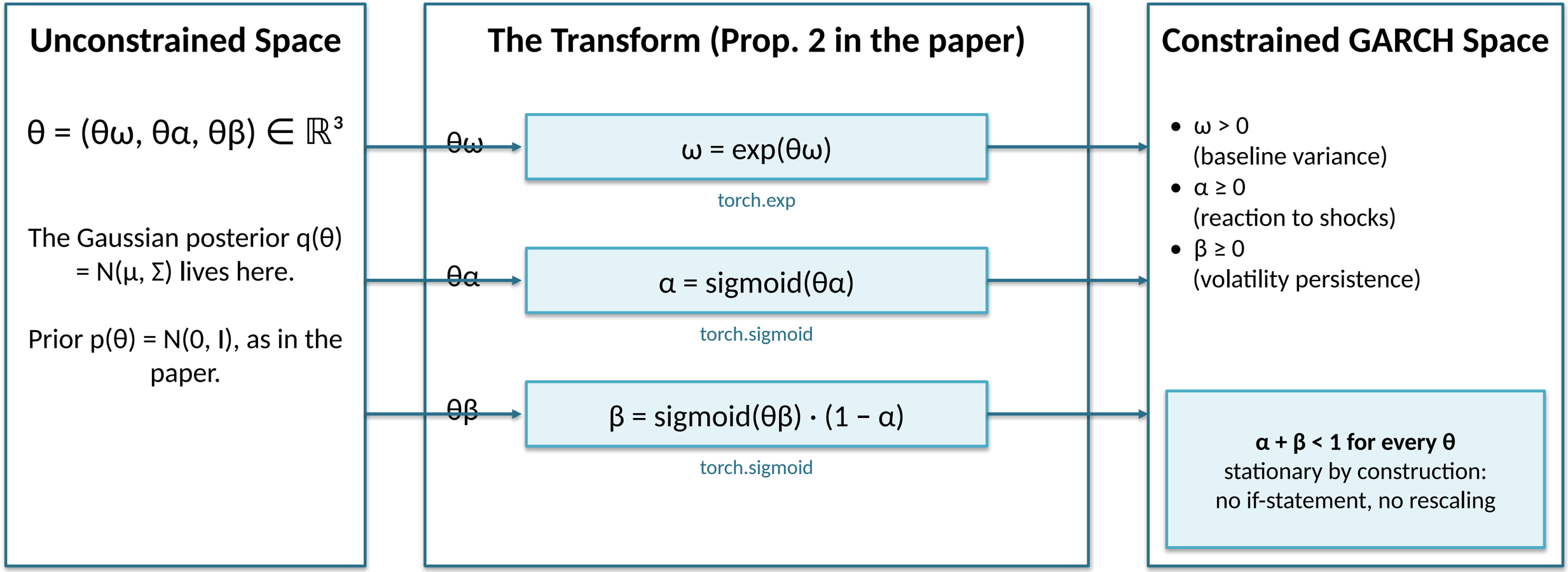

*The KL Divergence*

Maximizes the model's fit to the empirical financial data.

Monte Carlo: S = 50 (BBVI), S = 10 (full-cov).

distribution $\boldsymbol{q(\theta)}$ for deviating too far from standard prior $\boldsymbol{p(\theta)}$. Acts as

# Paper transforms map unconstrained VI parameters into a valid, stationary GARCH




**Fix vs. first draft:** the old "if α + β ≥ 0.99, rescale" rule was non-smooth and not the paper's method. The transform is smooth everywhere, so gradients are always well defined.

# The VI training loop: sample, score, update in PyTorch

## 1. Sample Parameters



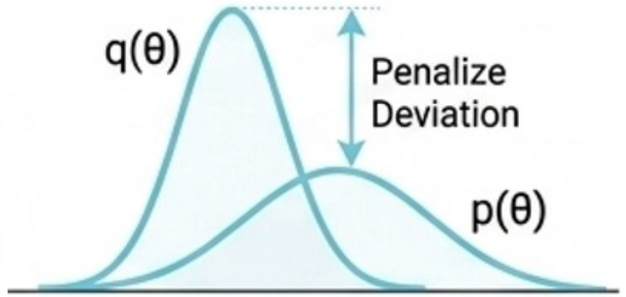
$$q(\theta) = N(\mu, \sigma^2)$$

## 2. Calculate Expected Log-Likelihood




Compute GARCH Likelihood over MC samples.

## 3. Compute KL Divergence




$$KL(q(\theta) \,||\, p(\theta))$$

## 4. Backpropagate ELBO Loss

loss.backward() → Adam step

Autodiff gradients

## 5. Update Parameters

Adam Optimizer Step (lr = 0.01)

Update $\mu, \sigma^2$

## The ELBO Objective

$$ELBO = \mathbb{E}[log\ p(data \mid \theta)] - KL(q(\theta) \,||\, p(\theta))$$

**Two gradient estimators**

**BBVI** (paper, Alg. 1): score-function gradient from likelihood values only. Diagonal q, S = 50, lr 0.005, momentum 0.4.

**Full-cov VI**: reparameterisation θ = μ + Lε, gradients by autodiff. Full Σ, S = 10, Adam lr 0.01.

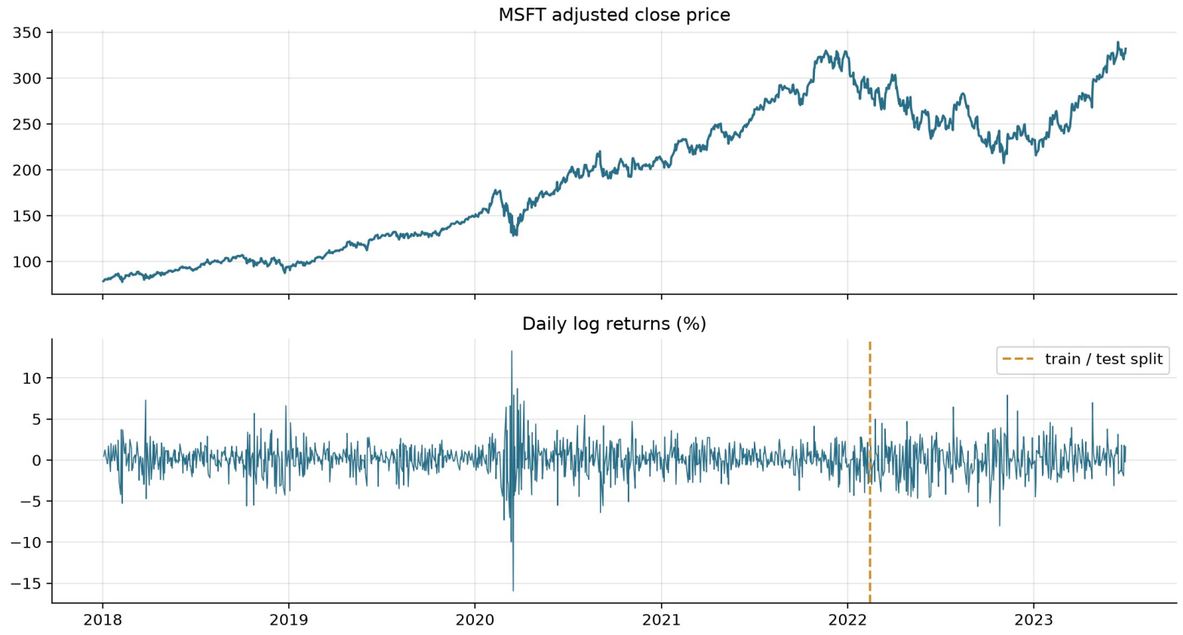
Same posterior target, prior and transforms.

# Data pipeline: the paper’s Microsoft case study, cleaned and split in time order

## Data Pipeline

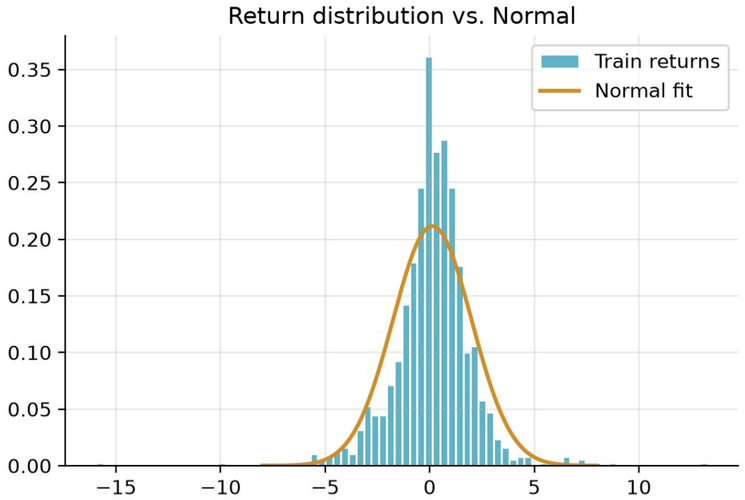
| Step | Detail |
|---|---|
| **Source** | Yahoo Finance (yfinance), adjusted close |
| **Asset** | Microsoft (MSFT), as in the paper; plus a Dow 30 panel |
| **Window** | 2 Jan 2018 – 30 Jun 2023: 1,383 days → 1,382 returns |
| **Cleaning** | 0 missing, 0 duplicates, 0 non-positive prices; longest gap 4 days (holidays) |
| **Returns** | $r_t = 100 \cdot (\log P_t - \log P_{t-1})$ |
| **Split** | 75% train (to 11 Feb 2022, 1,036 days) / 25% test (346 days) |

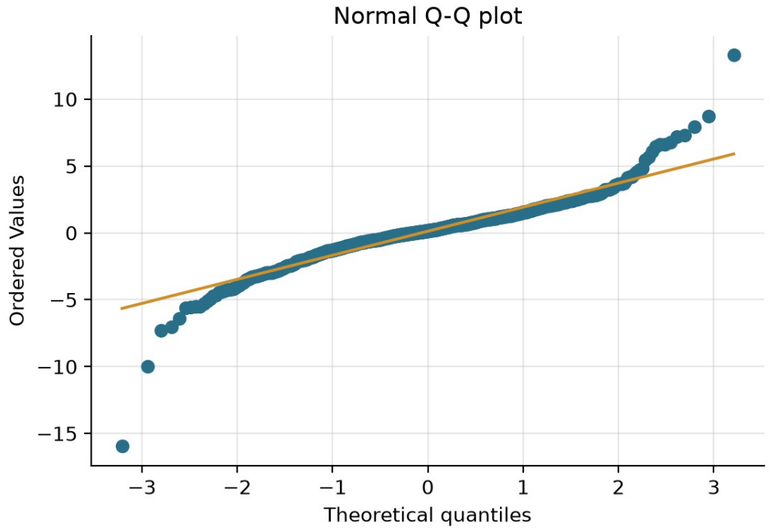
## Prices and Returns

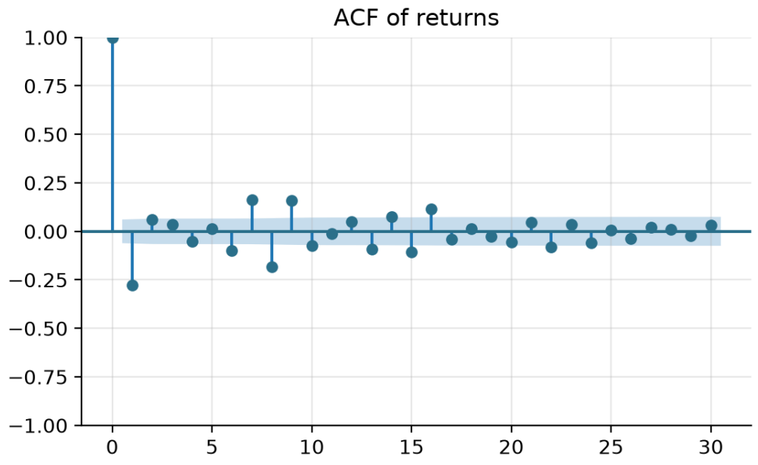


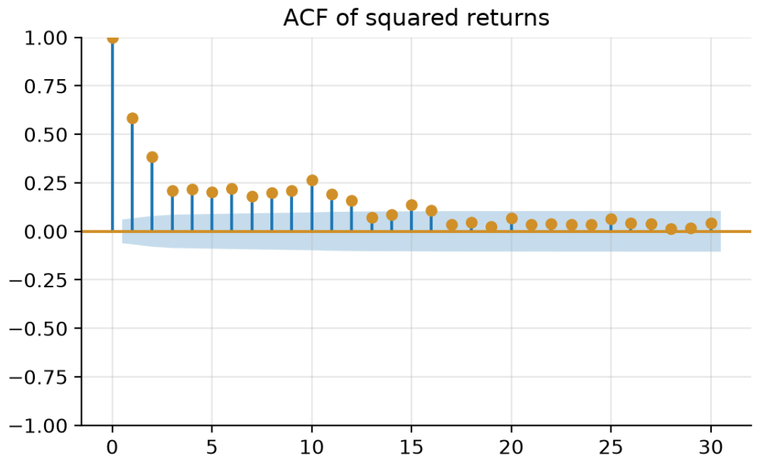

**Takeaway:** the test period (2022 rate hikes) is more volatile than most of the training period, so it is a genuine out-of-sample challenge.

# Exploratory analysis: MSFT returns show the stylised facts GARCH is built for




**Fat tails**
Excess kurtosis 9.5 (train); Jarque-Bera $p < 0.001$

**Volatility clustering**
Ljung-Box(10) on $r^2$: 890 ($p < 0.001$)

**ARCH effects**
Engle ARCH-LM(10): 393 ($p < 0.001$)

**Mean ≈ 0**
0.12%/day vs 1.9% daily volatility: under 0.5% of $E[r^2]$, so we use the paper's $\mu = 0$

**→ GARCH(1,1) for clustering; Student-t innovations for fat tails (extension)**

# Our GARCH(1,1) estimates replicate the paper's Table II for Microsoft

| Method | ω (ours) | α (ours) | β (ours) | ω (paper) | α (paper) | β (paper) |
|---|---|---|---|---|---|---|
| **QML** | 0.160 | 0.186 | 0.769 | 0.15 | 0.19 | 0.77 |
| **MCMC** | 0.217 ± 0.049 | 0.203 ± 0.031 | 0.731 ± 0.036 | 0.200 ± 0.046 | 0.203 ± 0.031 | 0.737 ± 0.036 |
| **BBVI (diagonal)** | 0.214 ± 0.022 | 0.201 ± 0.023 | 0.732 ± 0.023 | 0.196 ± 0.021 | 0.199 ± 0.023 | 0.740 ± 0.023 |
| **Full-cov VI*** | 0.220 ± 0.050 | 0.200 ± 0.029 | 0.732 ± 0.035 | 0.205 ± 0.047 | 0.201 ± 0.030 | 0.736 ± 0.035 |

Bayesian columns: posterior mean ± SD of transformed draws (paper eq. 6). *Paper column = MGVB, its full-covariance VI optimiser.

**Replicated**

Every estimate is within ≈ 0.02 of the paper; QML also matches the arch package.

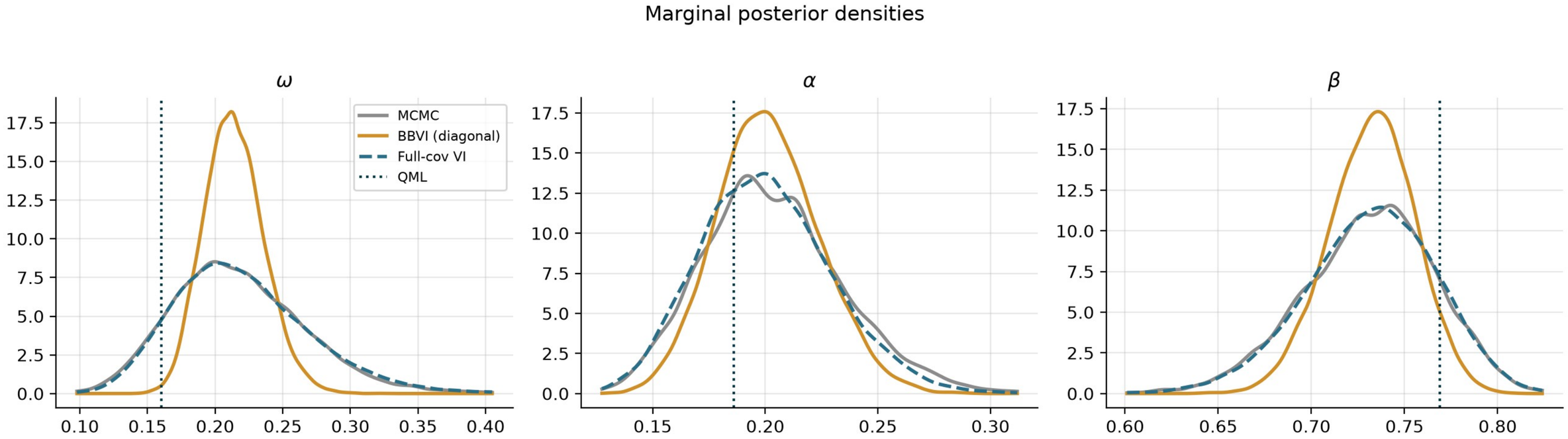
**Persistence**

$\alpha + \beta \approx 0.955$: a volatility shock has a half-life of ≈ 15 trading days.

**Out-of-sample check**

Test NLL (QML): 764.6 ours vs 761.4 in the paper.

# Full-covariance VI reproduces the MCMC posterior; diagonal BBVI is over-confident




**Posterior SD of ω**

MCMC: 0.049
Full-cov VI: 0.050
BBVI: 0.022 (−55%)

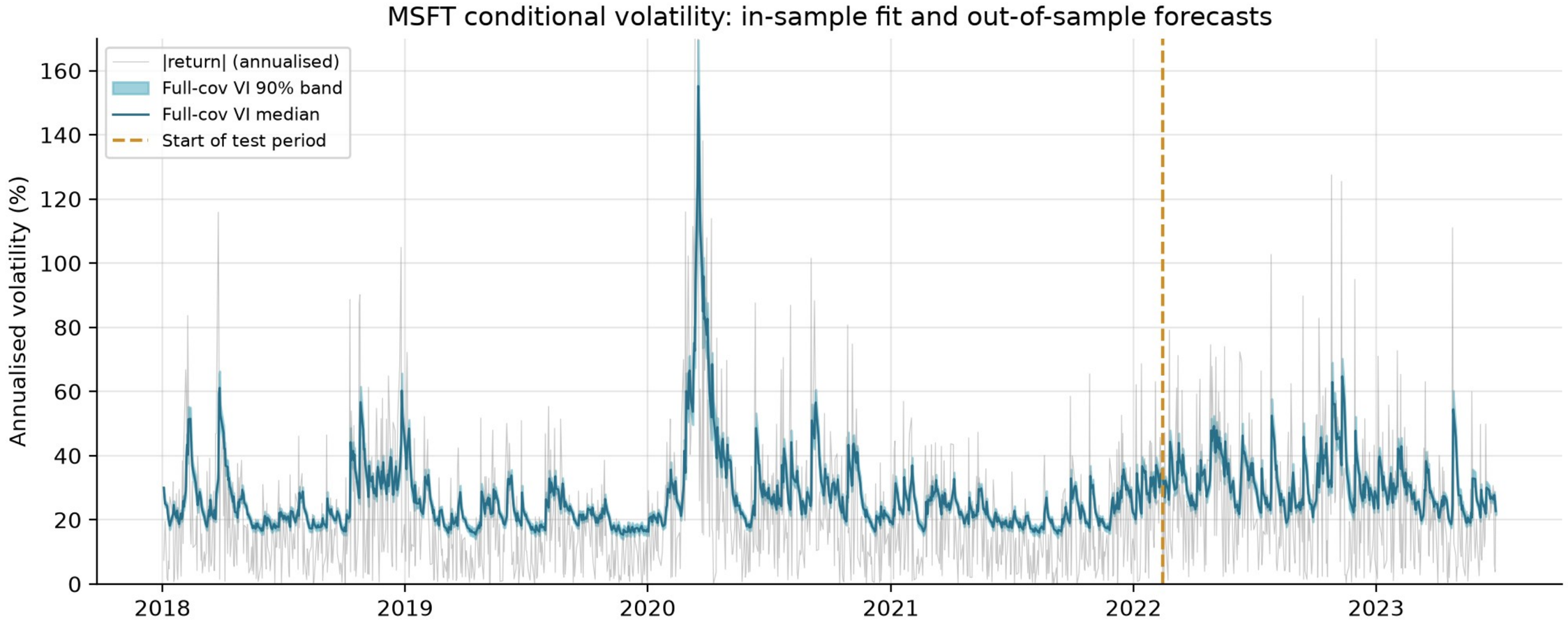
**Posterior SD of β**

MCMC: 0.036
Full-cov VI: 0.035
BBVI: 0.023 (−35%)

**Final ELBO (higher = closer)**

Full-cov VI: −1945.3
BBVI: −1946.2
Same ordering as the paper

# Bayesian estimation turns volatility into a forecast with credible bands




COVID-19 peak above 150% annualised; the 2022 rate-hike regime is tracked out of sample.
Average 90% band width (test): **MCMC 3.5 · Full-cov VI 3.5 · BBVI 3.1 vol-points** → diagonal VI is over-confident.

# VI forecasts as well as QML and MCMC, but it is not faster for a 3-parameter GARCH

## Out-of-sample accuracy: % difference vs QML

| Method | NLL | RMSE | MAD | QLIK |
|---|---|---|---|---|
| **MCMC** | +0.15 | +0.23 | −0.60 | +0.26 |
| **BBVI (diagonal)** | +0.19 | +0.18 | −0.82 | +0.32 |
| **Full-cov VI** | +0.16 | +0.13 | −0.94 | +0.27 |

All methods within 1% of QML (paper: sub-1% on average). Lower is better; negative = better than QML.

## Computational cost (MSFT, one run)

| Method | Run time | Likelihood evals |
|---|---|---|
| **QML** | < 0.01 s | ≈ 380 |
| **MCMC (RWM)** | ≈ 0.3 s | 30,000 |
| **BBVI (diagonal)** | ≈ 1 s | 125,000 |
| **Full-cov VI (PyTorch)** | ≈ 16 s | 15,000 |

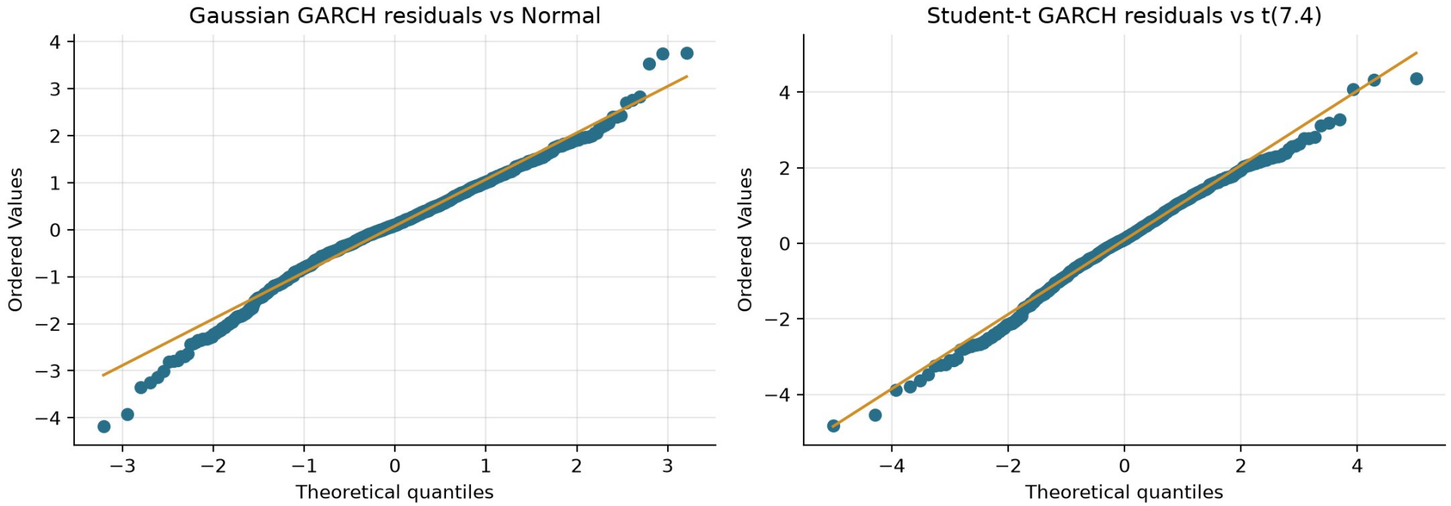
**Honest result:** with 3 parameters, a well-tuned MCMC is cheap and VI is not faster. (PyTorch time is mostly Python overhead in the recursion loop.)

**VI's real advantages:** one ELBO curve to monitor, the same hyperparameters for every stock, a reusable analytical posterior, and scalability to high-dimensional models where MCMC mixes slowly.

# Extension: Student-t innovations capture the fat tails that Gaussian GARCH misses

### Heavy Tails (Student-t Innovations)

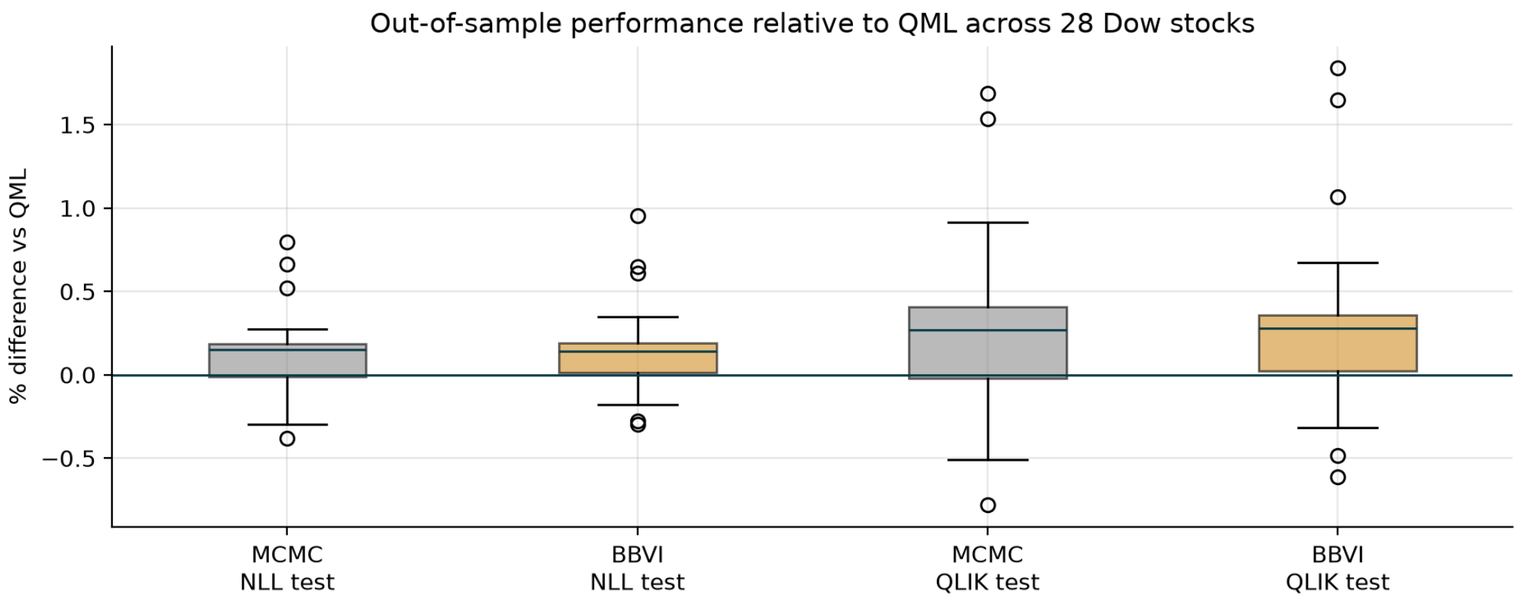
- **The problem:** Gaussian-GARCH residuals are still fat-tailed (left Q-Q).
- **The extension:** $\nu = 2 + \exp(\theta_\nu)$, estimated jointly with full-covariance VI (4 parameters).
- **The result:** $\nu \approx 7.4$ (90% interval 5.2–10.2), far from Normal.




| Criterion | Gaussian VI-GARCH | Student-t VI-GARCH |
|---|---|---|
| **ELBO (model evidence)** | −1945.3 | **−1934.6 (+10.7)** |
| **Test log-likelihood** | −765.9 | **−754.1 (+11.8)** |
| **Volatility path** | almost identical | almost identical |

# Scaling up: the same conclusion holds across 28 Dow Jones stocks

**Data:** 28 Dow 30 stocks (DOW, WBA dropped: incomplete) | 2018–2023 | **Split:** 75% / 25% | identical hyperparameters for every stock




### Mean % deviation vs QML

**Test NLL:** MCMC +0.12 · BBVI +0.14
**Test QLIK:** MCMC +0.25 · BBVI +0.28
**Train NLL:** QML lowest for 100% of stocks (it maximises it by design)

Average time per stock: MCMC ≈ 0.3 s, BBVI ≈ 1.1 s

### This project (28 stocks)

MCMC and BBVI forecasts sit within ≈ 0.3% of QML on average and are almost indistinguishable from each other.

### Magris & Iosifidis (488 stocks)

Across seven GARCH-family models and four VI optimisers, deviations from QML are sub-1% on average (paper, Table I).

# Execution Summary & Core Takeaways

**Replication**

- **Goal**: Implement VI for GARCH as in Magris & Iosifidis (2023).
- **Result**: QML, MCMC and BBVI match the paper's Table II within ≈ 0.02; the transforms guarantee $\alpha + \beta < 1$.

**Accuracy & Uncertainty**

- **Goal**: Match MCMC/QML accuracy.
- **Result**: VI is within 1% of QML out of sample (MSFT and 28 stocks). Full-cov VI ≈ MCMC posterior; diagonal BBVI is over-confident.

**Fat Tails & Cost**

- **Goal**: Model real-world returns efficiently.
- **Result**: Student-t ($\nu \approx 7$) wins on ELBO and test log-likelihood. At 3 parameters MCMC is cheap; VI pays off through generic tuning and scale.

Variational Inference is an accurate, well-calibrated alternative to MCMC for GARCH, as long as the variational family is rich enough (full covariance).

**Next steps:** GJR / EGARCH / FIGARCH · natural-gradient optimisers (QBVI, MGVB) · realised-volatility proxies from intraday data (e.g. WRDS TAQ)

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Patton, A. J. (2011). Volatility Forecast Comparison Using Imperfect Volatility Proxies. Journal of Econometrics, 160(1), 246–256.
Code & data: github.com/SGhuman123/Fordham-Time-Series-Econometrics (VI_GARCH.ipynb)